

SUDIPTA BASAK

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TECHNICAL SKILLS

Languages

C/C++, Java, Python, Kotlin, JavaScript, TypeScript, Rust, Solidity

Backend & Data

Spring Boot, Micronaut, REST APIs, GraphQL, Kafka, Redis, Spring Batch, PostgreSQL, Couchbase, DB2

Blockchain & ZK

Solidity, Foundry, Truffle/Hardhat, Circom & SnarkJS, ZoKrates, Waku, IPFS, Web3.py

Tools & Platforms

Git, Docker, VSCode, IntelliJ IDEA, PyCharm, GitHub Copilot, Gen AI, Linux, macOS, Windows

Foundations

Data Structures & Algorithms, OOP, Cryptography, Problem Solving

EDUCATION

IIT Jammu

M.Tech, Computer Science & Engineering
Minor: Information Security
CGPA 8.39 • 2024

UEM, Kolkata

B.Tech, Computer Science & Engineering
CGPA 8.50 • 2021

ACHIEVEMENTS

Aptos Winter School & Hackathon

Built *CrysPay* (crypto transfers via phone numbers); 3rd of 5 finalists; \$5,000 grant from the Aptos Foundation.

Amex GrowthHack'25

Top-5 finalist team out of 500+ worldwide.

Freelance Blockchain Developer

Clients across India, the US, the UK, and Dubai.

PUBLICATIONS

Privacy-Preserving Location-Based Services: A DQN Algorithmic Perspective [Springer]

On the Design of Ethereum Data Availability Sampling: A Comprehensive Simulation Study [arXiv]

EXPERIENCE

American Express

Aug 2024 – Present

Software Developer I

Gurgaon, India

- Migrated card privacy-preference data from legacy Mainframe/DB2 to a modern Couchbase, Java, and Spring Boot stack.
- Modernized internal Lookup Tables, transitioning from Mainframe/DB2 to Java and GraphQL-based services.
- Built data-processing workflows leveraging Redis, Kafka, Spring Batch, and PostgreSQL.
- Improved code quality and observability via structured logging of events and exceptions across applications.

Nethermind

Jun 2024 – Aug 2024

Blockchain Developer & Researcher Intern

London, UK (Remote)

- Co-developed FRIDA, a FRI-commitment-scheme-based Data Availability (DA) layer for blockchains.
- Built a Rust CLI to drive the FRIDA proof-of-concept and authored Rust test suites validating it.

Status

Feb 2024 – Aug 2024

R&D Engineer Intern

Zug, Switzerland (Remote)

- Contributed to a Data Availability Sampling (DAS) research project in collaboration with the Ethereum Foundation.
- Developed a Python simulator modeling large-scale distributed Ethereum nodes to analyze data sharding.

Pegasystems

Jun 2023 – Dec 2023

Software Engineer Intern

Hyderabad, India

- Integrated Meta's open-source LLaMA 2 LLM with Pega Launchpad to assist users in authoring expressions.
- Curated an expression-examples dataset and applied vector embeddings to fine-tune the LLaMA 2 13B model, improving domain-specific accuracy.
- Exposed the model through a Kotlin (Micronaut) REST API and containerized the application with Docker.

PROJECTS

WhisUp3 — Decentralised Anonymous Feedback



Jan 2024 – Feb 2024

ReactJS, TypeScript, Waku, MetaMask

- Built a decentralised anonymous feedback platform on the Waku communication protocol.
- Enabled anyone to receive encrypted, anonymous messages on their public Ethereum address.
- Secured messages with public-key encryption and MetaMask-based private-key decryption.

ZK Authentication



Dec 2023 – Feb 2024

Solidity, Groth16, ZoKrates

- Implemented zkSNARK (Groth16) on-chain authentication for EVM smart contracts.
- Used the ZoKrates toolbox and zokrates.js for proof generation and verification.
- Authenticated users via zero-knowledge proof of knowledge of a hash preimage.

Wild-Storage



Feb 2023 – Jan 2024

Python, C++, Docker

- Engineered a tool using YouTube as unlimited file storage by encoding file binaries as black/white pixels and decoding videos to restore files.
- Devised an encoding scheme resilient to YouTube's video compression to prevent data loss.
- Ported hot paths to C++ (wrapped in Python) for a 3x speedup; shipped a CLI and Dockerfile.

PancakeSwap Sniping Bot



2021

Python, Web3.py, Telethon

- Built a bot to automate cryptocurrency trading on EVM-based AMMs, sourcing Telegram signals via Telethon and executing on-chain trades through Web3.py.